CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 10: Arithmetic Progression

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Find the common difference and the 10th term of the A.P. $11, 7, 3, \ldots$
2. Which term of the A.P. $2, 9, 16, \ldots$ is 149?
3. Is 68 a term of the A.P. $7, 10, 13, \ldots$?
4. The 3rd term of an A.P. is 12 and the 8th term is 37. Find the 20th term.
5. Find the general term of the A.P. $9, 14, 19, \ldots$
6. Find the sum of the first 16 terms of the A.P. $10, 6, 2, \ldots$
7. Find the sum of all natural numbers between 100 and 200 that are divisible by 9.
8. The sum of the first n terms of an A.P. is $2n^2 + 3n$. Find its 10th term.
9. Find the arithmetic mean of 14 and 38, and the value of x if $x + 2$, $3x$, $4x + 1$ are in A.P.
10. The sum of three numbers in A.P. is 30 and the sum of their squares is 318. Find the numbers.

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Answer Key

1. $d = -4$; $t_{10} = 11 - 36 = -25$
2. The 22nd term
3. $n - 1 = \frac{61}{3}$, not a whole number; no
4. $d = 5$, $a = 2$; $t_{20} = 97$
5. $t_n = 5n + 4$
6. $8\,[\,20 - 60\,] = -320$
7. $108 + 117 + \cdots + 198$: 11 terms, sum $= 1683$
8. $S_{10} - S_9 = 230 - 189 = 41$
9. 26; $6x = 5x + 3$, so $x = 3$
10. $a = 10$, $2d^2 = 18$; 7, 10, 13

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